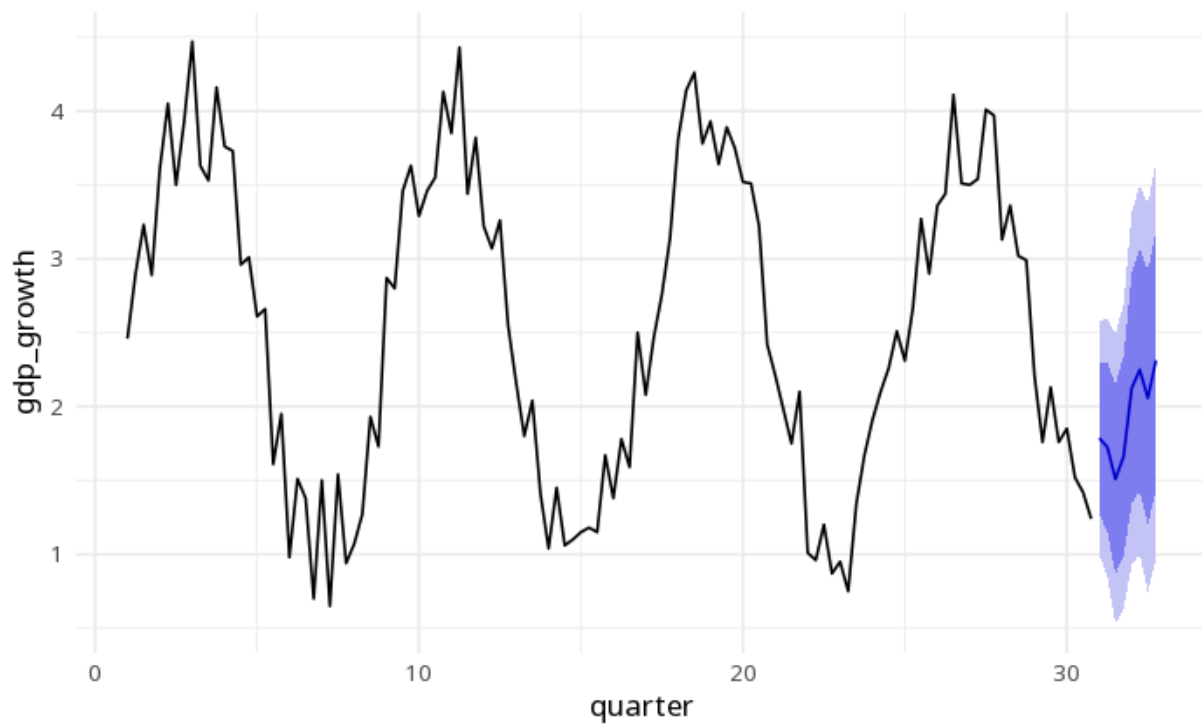
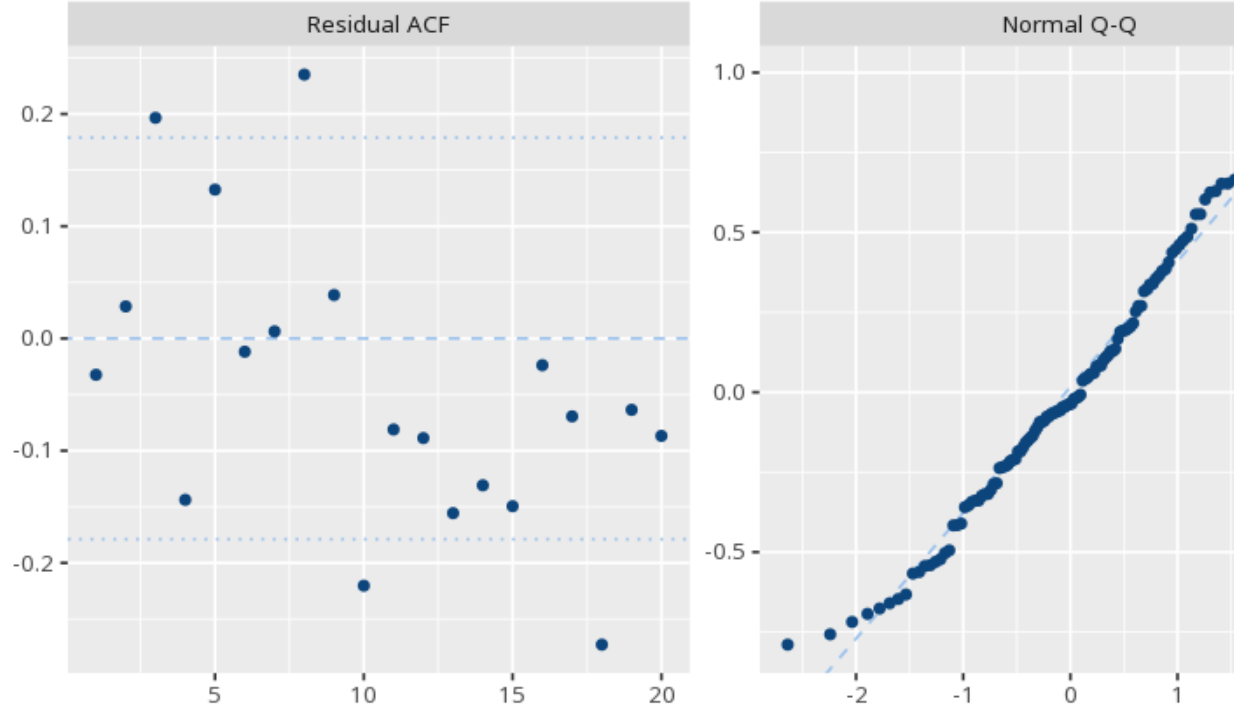


# 1 ARIMA / SARIMA

|                                                                     |                                   |              |              |
|---------------------------------------------------------------------|-----------------------------------|--------------|--------------|
|                                                                     | (1)                               |              |              |
| ar1                                                                 | 0.48<br>(0.11)<br>[0.27, 0.68]    |              |              |
| ar2                                                                 | 0.33<br>(0.09)<br>[0.16, 0.50]    |              |              |
| sar1                                                                | 1.06<br>(0.12)<br>[0.81, 1.30]    |              |              |
| sar2                                                                | −0.53<br>(0.12)<br>[−0.76, −0.30] |              |              |
| sma1                                                                | −0.70<br>(0.08)<br>[−0.87, −0.54] |              |              |
| intercept                                                           | 2.49<br>(0.12)<br>[2.25, 2.73]    |              |              |
| Num.Obs.                                                            | 120                               |              |              |
| $\sigma^2$                                                          | 0.16                              |              |              |
| Ljung–Box p                                                         | <.001                             |              |              |
| AIC                                                                 | 133.95                            |              |              |
| BIC                                                                 | 153.46                            |              |              |
| Log.Lik.                                                            | −59.98                            |              |              |
| Ljung–Box Q(5) = 23.82, p < .001 (lag 10, residual autocorrelation) |                                   |              |              |
| Period                                                              | Forecast                          | 80% PI       | 95% PI       |
| 1                                                                   | 1.79                              | [1.27, 2.30] | [1.00, 2.58] |
| 2                                                                   | 1.73                              | [1.15, 2.30] | [0.85, 2.60] |
| 3                                                                   | 1.51                              | [0.87, 2.15] | [0.53, 2.49] |
| 4                                                                   | 1.66                              | [0.98, 2.34] | [0.63, 2.70] |
| 5                                                                   | 2.12                              | [1.34, 2.90] | [0.93, 3.31] |
| 6                                                                   | 2.25                              | [1.43, 3.07] | [1.00, 3.50] |
| 7                                                                   | 2.06                              | [1.20, 2.91] | [0.75, 3.37] |
| 8                                                                   | 2.31                              | [1.43, 3.19] | [0.97, 3.66] |

Forecasts from ARIMA(2,0,0)(2,0,1)[4] with non-zero mean





The chosen  $(p,d,q)(P,D,Q)$  orders describe the model. The Ljung–Box  $Q(df)$  row tests whether the residuals still carry autocorrelation up to the reported lag (a large  $p$  means none is detected); use it with AIC/BIC to judge fit. The forecast table and plot give predictions with uncertainty bands. AIC/BIC compare models only when fit to the same series with identical differencing ( $d, D$ ) and transformation; lower is better, but they are relative, not absolute, measures of fit. Method: R forecast package (Hyndman & Khandakar, 2008; Hyndman & Athanasopoulos, Forecasting: Principles and Practice).

An  $ARIMA(2,0,0)(2,0,1[4])$  model was fit (AIC=133.95); the Ljung–Box test of residual autocorrelation gave  $p < .001$ .

Statistical basis: ARIMA/SARIMA model selection and forecasting (Hyndman R, Athanasopoulos G, Bergmeir C, et al. (2024). forecast: Forecasting functions for time series and linear models. R package.).